Data Dictionary

Rockbuster Stealth LLC

Author: Helena Freitas

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1. Introduction

This **Data Dictionary** serves as reference for the Rockbuster Stealth relational database. It is designed to help analysts and developers understand the context, structure, and relationships of the data used for business intelligence and reporting.

Rockbuster's data is organized into a **Snowflake Schema**-like structure, since dimension tables are surrounded by sub-dimension tables (for example: staff → address → city → country).

- **Fact Tables:** Tables like payment and rental record the core business transactions (the "actions").
- **Dimension Tables:** Tables such as film, customer, and address provide the descriptive context (the "who, what, and where").

2. Glossary

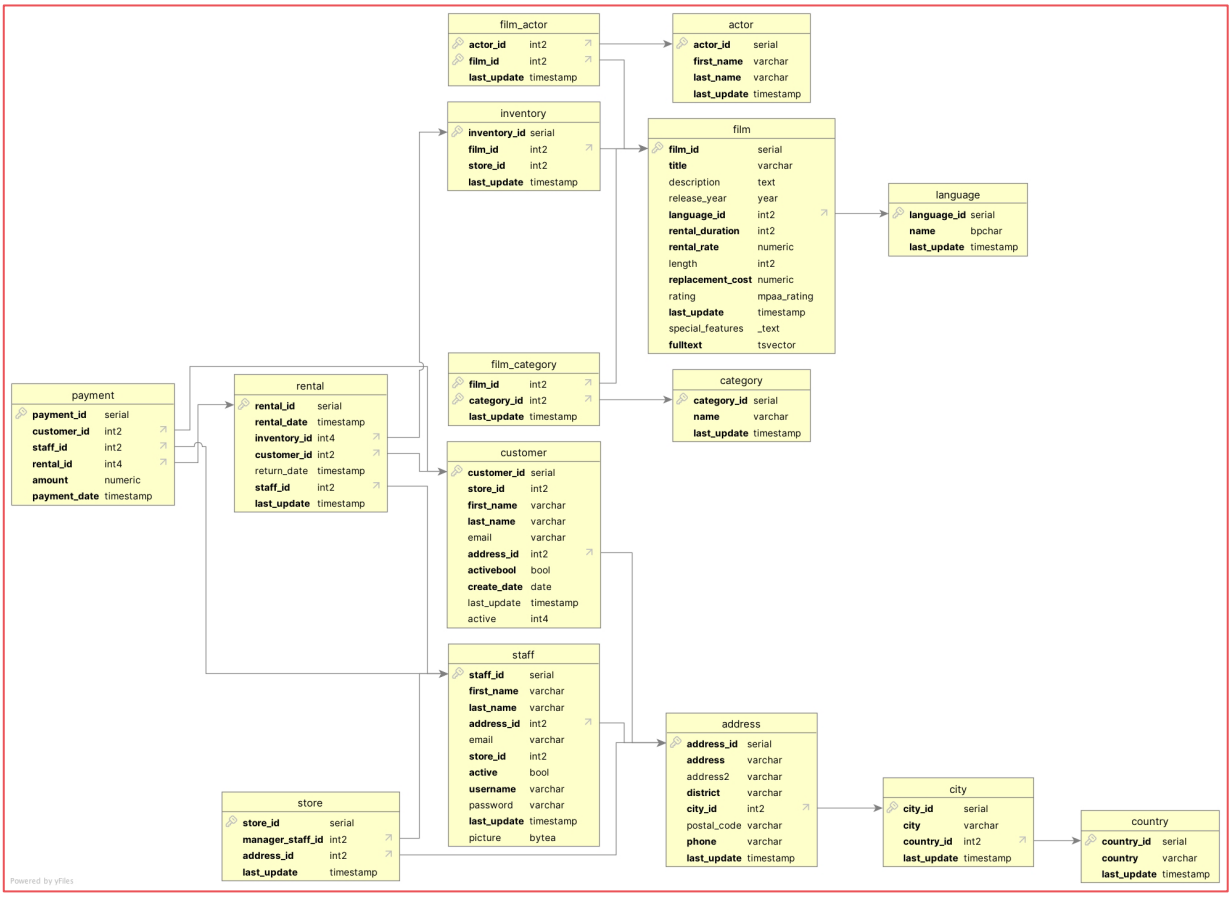
To navigate this document effectively, please refer to the following definitions:

- **Primary Key (PK):** A column that uniquely identifies every row in a table. It cannot contain duplicate values or empty cells. It is the "anchor" of the table.
- **Foreign Key (FK):** A column that creates a link between two tables. It "points" to the Primary Key of another table, ensuring that their relationship remains valid.
- **Composite Key:** A key that uses two or more columns combined to create a unique identifier. This is common where a single ID isn't enough to guarantee uniqueness.

This dictionary includes "Links to" and "Links from" columns to visualize data flow:

- Links to: Identifies which transaction or fact tables use this table for context.
- Links from: Identifies where the current table gets its descriptive data.

3. Entity Relationship Diagram (ERD)



The ERD works as the relational map. Its primary purpose is to guide the user on which tables must be joined together to get a complete answer to a business question.

4. Fact tables

4.1. Payment table

Column Name	Data Type	Description	Relatsh.
payment_id	integer	Unique payment identifier	PK
customer_id	smallint	Paying customer	FK
staff_id	smallint	Staff member processing payment	FK
rental_id	integer	Related rental transaction	FK
amount	numeric	Payment amount	
payment_date	timestamp without time zone	Date of payment	

Links to:

- customer_id → Customer table
- staff_id → Staff table
- rental_id → Rental table

Links from:

None

Unique key: payment_id

4.2. Rental table

Column Name	Data Type	Description	Relatsh.
rental_id	integer	Unique rental transaction identifier	PK
rental_date	timestamp without time zone	Date the rental occurred	
inventory_id	integer	Rented physical copy	FK
customer_id	smallint	Customer who rented	FK
return_date	timestamp without time zone	Date item was returned	
staff_id	smallint	Staff member handling rental	FK
last_update	integer	Last modification timestamp	

Links to:

- inventory_id → Inventory table
- customer_id → Customer table
- staff_id → Staff table

Links from:

- Referenced by Payment table (rental_id)

Unique key: rental_id

5. Dimension tables

5.1. Customer table

Column Name	Data Type	Description	Relatsh.
customer_id	integer	Unique customer identifier	PK
store_id	smallint	Store the customer belongs to	
first_name	character varying	Customer first name	
last_name	character varying	Customer last name	
email	character varying	Customer email address	
address_id	smallint	Customer address	FK
activebool	boolean	Whether customer is active	
create_date	date	Date customer was created	
last_update	timestamp without time zone	Last modification timestamp	
active	integer	Active status	

Links to:

- address_id → Address table
- store_id → Store table

Links from:

- Referenced by Payment table (customer_id)
- Referenced by Rental table (customer_id)

Unique key: customer_id

5.2. Staff table

Column Name	Data Type	Description	Relatsh.
staff_id	integer	Unique staff identifier	PK
first_name	character varying	Staff first name	
last_name	character varying	Staff last name	
address_id	smallint	Staff address	FK
email	character varying	Staff email	
store_id	smallint	Store where staff works	
active	boolean	Active employment status	
username	character varying	Login username	
password	character varying	Login password	
last_update	timestamp without time zone	Last modification timestamp	
picture	bytea	Staff profile picture	

Links to:

- address_id → Address table
- store_id → Store table

Links from:

- Referenced by Payment table (staff_id)
- Referenced by Rental table (staff_id)
- Referenced by Store table (manager_staff_id)

Unique key: staff_id

5.3. Store table

Column Name	Data Type	Description	Relatsh.
store_id	integer	Unique store identifier	PK
manager_staff_id	smallint	Store manager	FK
address_id	smallint	Store address	FK
last_update	timestamp without time zone	Last modification timestamp	

Links to:

- address_id → Address table
- manager_staff_id → Staff table

Links from:

- Referenced by Inventory table (store_id)
- Referenced by Customer table (store_id)
- Referenced by Staff table (store_id)

Unique key: store_id

5.4. Inventory table

Column Name	Data Type	Description	Relatsh.
inventory_id	integer	Unique inventory identifier	PK
film_id	smallint	Film in inventory	FK
store_id	smallint	Store holding the inventory	
last_update	timestamp without time zone	Last modification timestamp	

Links to:

- film_id → Film table
- store_id → Store table

Links from:

- Referenced by Rental table (inventory_id)

Unique key: inventory_id

5.5. Film table

Column Name	Data Type	Description	Relatsh.
film_id	integer	Unique film identifier	PK
title	character varying	Film title	
description	text	Film description	
release_year	integer	Film release year	
language_id	smallint	Film language	FK
rental_duration	smallint	Rental duration	
rental_rate	numeric	Rental price	
length	smallint	Film length	
replacement_cost	numeric	Cost to replace film	
rating	USER-DEFINED	Film rating	
last_update	timestamp without time zone	Last modification timestamp	

special_features	ARRAY	List of special features	
fulltext	tsvector	Full-text search	

Links to:

- language_id → Language table

Links from:

- Referenced by Inventory table (film_id)
- Referenced by Film_actor table (film_id)
- Referenced by Film_category table (film_id)

Unique key: film_id

5.6. Actor table

Column Name	Data Type	Description	Relatsh.
actor_id	integer	Unique actor identifier	PK
first_name	character varying	Actor first name	
last_name	character varying	Actor last name	
last_update	timestamp without time zone	Last modification timestamp	

Links to:

None

Links from:

- Referenced by Film_actor table (actor_id)

Unique key: actor_id

5.7. Category table

Column Name	Data Type	Description	Relatsh.
category_id	integer	Unique category identifier	PK
name	character varying	Category name	
last_update	timestamp without time zone	Last modification timestamp	

Links to:

None

Links from:

- Referenced by Film_category table (category_id)

Unique key: category_id

5.8. Language table

Column Name	Data Type	Description	Relatsh.
language_id	integer	Unique language identifier	PK
name	character	Language name	
last_update	timestamp without time zone	Last modification timestamp	

Links to:

None

Links from:

- Referenced by Film table (language_id)

Unique key: language_id

5.9. Address table

Column Name	Data Type	Description	Relatsh.
address_id	integer	Unique address identifier	PK
address	character varying	Street address	
address2	character varying	Additional address information	
district	character varying	District or region	
city_id	smallint	Unique city identifier	FK
postal_code	character varying	Postal code	
phone	character varying	Contact phone number	
last_update	timestamp without time zone	Last modification timestamp	

Links to:

- city_id → City table

Links from:

- Referenced by Customer table (address_id)
- Referenced by Staff table (address_id)
- Referenced by Store table (address_id)

Unique key: address_id

5.10. City table

Column Name	Data Type	Description	Relatsh.
city_id	integer	Unique city identifier	PK
city	character varying	City name	
country_id	smallint	Country	FK
last_update	timestamp without time zone	Last modification timestamp	

Links to:

- country_id → Country table

Links from:

- Referenced by Address table (city_id)

Unique key: city_id

5.11. Country table

Column Name	Data Type	Description	Relatsh.
country_id	integer	Unique country identifier	PK
country	character varying	Country name	
last_update	timestamp without time zone	Last modification timestamp	

Links to:

None

Links from:

- Referenced by City table (country_id)

Unique key: country_id

5.12. Film_actor table

Column Name	Data Type	Description	Relatsh.
actor_id	smallint	Actor identifier	PK
film_id	smallint	Film identifier	PK
last_update	timestamp without time zone	Last modification timestamp	

Links to:

- film_id → Film table
- actor_id → Actor table

Links from:

None

Composite key: actor_id, film_id

5.13. Film_category table

Column Name	Data Type	Description	Relatsh.
film_id	smallint	Film identifier	FK
category_id	smallint	Category identifier	FK
last_update	timestamp without time zone	Last modification timestamp	

Links to:

- film_id → Film table
- category_id → Category table

Links from:

None

Composite key: film_id, category_id